

# GraphTheorySymbol Memo

## v1.8.0 2026/09/28

Add in your preamble: `\usepackage{graph-theory-symbol}`.

For clarity, below we only show long macro names and short macro names; see Detailed usage section for intermediate length macro names.

| Long name                                              | Short name              | Result  | In context  |
|--------------------------------------------------------|-------------------------|---------|-------------|
| <code>\GTSadjacency/</code>                            | <code>\GTSa/</code>     | $\circ$ | $u \circ v$ |
| <code>\GTSadjacencyRelation/</code>                    | <code>\GTSaR/</code>    | $\circ$ | $u \circ v$ |
| <code>\GTSunadjacency/</code>                          | <code>\GTSu/</code>     | $\circ$ | $u \circ v$ |
| <code>\GTSunadjacencyRelation/</code>                  | <code>\GTSuR/</code>    | $\circ$ | $u \circ v$ |
| <code>\GTSdirectedAdjacencyUp/</code>                  | <code>\GTSdAU/</code>   | $\circ$ | $u \circ v$ |
| <code>\GTSdirectedAdjacencyUpRelation/</code>          | <code>\GTSdAUR/</code>  | $\circ$ | $u \circ v$ |
| <code>\GTSnegatedDirectedAdjacencyUp/</code>           | <code>\GTSnDAU/</code>  | $\circ$ | $u \circ v$ |
| <code>\GTSnegatedDirectedAdjacencyUpRelation/</code>   | <code>\GTSnDAUR/</code> | $\circ$ | $u \circ v$ |
| <code>\GTSdirectedAdjacencyDown/</code>                | <code>\GTSdAD/</code>   | $\circ$ | $u \circ v$ |
| <code>\GTSdirectedAdjacencyDownRelation/</code>        | <code>\GTSdADR/</code>  | $\circ$ | $u \circ v$ |
| <code>\GTSnegatedDirectedAdjacencyDown/</code>         | <code>\GTSnDAD/</code>  | $\circ$ | $u \circ v$ |
| <code>\GTSnegatedDirectedAdjacencyDownRelation/</code> | <code>\GTSnDADR/</code> | $\circ$ | $u \circ v$ |
| <code>\GTSincidence/</code>                            | <code>\GTSi/</code>     | $\circ$ | $v \circ e$ |
| <code>\GTSincidenceRelation/</code>                    | <code>\GTSiR/</code>    | $\circ$ | $v \circ e$ |
| <code>\GTSnegatedIncidence/</code>                     | <code>\GTSnI/</code>    | $\circ$ | $v \circ e$ |
| <code>\GTSnegatedIncidenceRelation/</code>             | <code>\GTSnIR/</code>   | $\circ$ | $v \circ e$ |
| <code>\GTSoutIncidence/</code>                         | <code>\GTSOI/</code>    | $\circ$ | $v \circ e$ |
| <code>\GTSoutIncidenceRelation/</code>                 | <code>\GTSOIR/</code>   | $\circ$ | $v \circ e$ |
| <code>\GTSnegatedOutIncidence/</code>                  | <code>\GTSnOI/</code>   | $\circ$ | $v \circ e$ |
| <code>\GTSnegatedOutIncidenceRelation/</code>          | <code>\GTSnOIR/</code>  | $\circ$ | $v \circ e$ |
| <code>\GTSinIncidence/</code>                          | <code>\GTSiI/</code>    | $\circ$ | $v \circ e$ |
| <code>\GTSinIncidenceRelation/</code>                  | <code>\GTSiIR/</code>   | $\circ$ | $v \circ e$ |
| <code>\GTSnegatedInIncidence/</code>                   | <code>\GTSnII/</code>   | $\circ$ | $v \circ e$ |
| <code>\GTSnegatedInIncidenceRelation/</code>           | <code>\GTSnIIR/</code>  | $\circ$ | $v \circ e$ |

| Long name                                           | Short name  | Result | In context |
|-----------------------------------------------------|-------------|--------|------------|
| \GTSadjacencyMonochrome/                            | \GTSaM/     |        | u  v       |
| \GTSadjacencyMonochromeRelation/                    | \GTSaMR/    |        | u  v       |
| \GTSunadjacencyMonochrome/                          | \GTSuM/     |        | u  v       |
| \GTSunadjacencyMonochromeRelation/                  | \GTSuMR/    |        | u  v       |
| \GTSdirectedAdjacencyUpMonochrome/                  | \GTSdAUM/   |        | u  v       |
| \GTSdirectedAdjacencyUpMonochromeRelation/          | \GTSdAUMR/  |        | u  v       |
| \GTSnegatedDirectedAdjacencyUpMonochrome/           | \GTSnDAUM/  |        | u  v       |
| \GTSnegatedDirectedAdjacencyUpMonochromeRelation/   | \GTSnDAUMR/ |        | u  v       |
| \GTSdirectedAdjacencyDownMonochrome/                | \GTSdADM/   |        | u  v       |
| \GTSdirectedAdjacencyDownMonochromeRelation/        | \GTSdADMR/  |        | u  v       |
| \GTSnegatedDirectedAdjacencyDownMonochrome/         | \GTSnDADM/  |        | u  v       |
| \GTSnegatedDirectedAdjacencyDownMonochromeRelation/ | \GTSnDADMR/ |        | u  v       |
| \GTSincidenceMonochrome/                            | \GTSiM/     |        | v  e       |
| \GTSincidenceMonochromeRelation/                    | \GTSiMR/    |        | v  e       |
| \GTSnegatedIncidenceMonochrome/                     | \GTSnIM/    |        | v  e       |
| \GTSnegatedIncidenceMonochromeRelation/             | \GTSnIMR/   |        | v  e       |
| \GTSoutIncidenceMonochrome/                         | \GTSoIM/    |        | v  e       |
| \GTSoutIncidenceMonochromeRelation/                 | \GTSoIMR/   |        | v  e       |
| \GTSnegatedOutIncidenceMonochrome/                  | \GTSnOIM/   |        | v  e       |
| \GTSnegatedOutIncidenceMonochromeRelation/          | \GTSnOIMR/  |        | v  e       |
| \GTSinIncidenceMonochrome/                          | \GTSiIM/    |        | v  e       |
| \GTSinIncidenceMonochromeRelation/                  | \GTSiIMR/   |        | v  e       |
| \GTSnegatedInIncidenceMonochrome/                   | \GTSnIIM/   |        | v  e       |
| \GTSnegatedInIncidenceMonochromeRelation/           | \GTSnIIMR/  |        | v  e       |

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| \GTSsetMainColor{blue}                    |            |        |            |
|-------------------------------------------|------------|--------|------------|
| Long name                                 | Short name | Result | In context |
| \GTSadjacency/                            | \GTSa/     |        | u  v       |
| \GTSadjacencyRelation/                    | \GTSaR/    |        | u  v       |
| \GTSunadjacency/                          | \GTSu/     |        | u  v       |
| \GTSunadjacencyRelation/                  | \GTSuR/    |        | u  v       |
| \GTSdirectedAdjacencyUp/                  | \GTSdAU/   |        | u  v       |
| \GTSdirectedAdjacencyUpRelation/          | \GTSdAUR/  |        | u  v       |
| \GTSnegatedDirectedAdjacencyUp/           | \GTSnDAU/  |        | u  v       |
| \GTSnegatedDirectedAdjacencyUpRelation/   | \GTSnDAUR/ |        | u  v       |
| \GTSdirectedAdjacencyDown/                | \GTSdAD/   |        | u  v       |
| \GTSdirectedAdjacencyDownRelation/        | \GTSdADR/  |        | u  v       |
| \GTSnegatedDirectedAdjacencyDown/         | \GTSnDAD/  |        | u  v       |
| \GTSnegatedDirectedAdjacencyDownRelation/ | \GTSnDADR/ |        | u  v       |
| \GTSincidence/                            | \GTSi/     |        | v  e       |
| \GTSincidenceRelation/                    | \GTSiR/    |        | v  e       |
| \GTSnegatedIncidence/                     | \GTSnI/    |        | v  e       |
| \GTSnegatedIncidenceRelation/             | \GTSnIR/   |        | v  e       |
| \GTSoutIncidence/                         | \GTSoI/    |        | v  e       |
| \GTSoutIncidenceRelation/                 | \GTSoIR/   |        | v  e       |
| \GTSnegatedOutIncidence/                  | \GTSnOI/   |        | v  e       |
| \GTSnegatedOutIncidenceRelation/          | \GTSnOIR/  |        | v  e       |
| \GTSinIncidence/                          | \GTSiI/    |        | v  e       |
| \GTSinIncidenceRelation/                  | \GTSiIR/   |        | v  e       |
| \GTSnegatedInIncidence/                   | \GTSnII/   |        | v  e       |
| \GTSnegatedInIncidenceRelation/           | \GTSnIIR/  |        | v  e       |

| Long name                 | Short name           | Result         | In context             |
|---------------------------|----------------------|----------------|------------------------|
| <code>\GTSmetaNot/</code> | <code>\GTSmN/</code> | $\overline{m}$ | $\overline{m}^{\circ}$ |
| <code>\GTSmetaAnd/</code> | <code>\GTSmA/</code> | $\wedge^m$     | $\wedge^{\circ m}$     |
| <code>\GTSmetaOr/</code>  | <code>\GTSmO/</code> | $\vee^m$       | $\vee^{\circ m}$       |

Meta-logic can be used for compound adjacency types, etc.

Implicit meta-and via concatenation, no space except around whole relations compositions, no parentheses via higher priority to meta-and, brackets when all possible adjacencies are specified between two vertices:

Tournament adjacency:

$\forall u, v, u [\overline{m}^{\circ} \vee^{\circ m} v]$ .

Tournament adjacency with maybe some colored edge also, who knows?:

$\forall u, v, u [\overline{m}^{\circ} \vee^{\circ m} v]$ .

Directed graph:

$\forall u, v, u [\overline{m}^{\circ} \vee^{\circ m} \vee^{\circ m} \vee^{\circ m} v]$ .

Oriented graph:

$\forall u, v, u [\overline{m}^{\circ} \vee^{\circ m} \vee^{\circ m} v]$ .

Tournament adjacency:

$[\overline{m}^{\circ} \vee^{\circ m}]$  is obtained with `\mathrel{\{[\backslashGTSdAD/\backslashGTSnDAU/\backslashGTSmO/\backslashGTSnDAD/\backslashGTSdAU/]\}}`.

Tournament adjacency with maybe some colored edge also, who knows?:

$\overline{m}^{\circ} \vee^{\circ m}$  is obtained with `\mathrel{\{[\backslashGTSdAD/\backslashGTSnDAU/\backslashGTSmO/\backslashGTSnDAD/\backslashGTSdAU/]\}}`.

Directed graph:

$[\overline{m}^{\circ} \vee^{\circ m} \vee^{\circ m} \vee^{\circ m}]$  is obtained with

`\mathrel{\{[\backslashGTSdAD/\backslashGTSnDAU/\backslashGTSmO/\backslashGTSnDAD/\backslashGTSdAU/\backslashGTSmO/\backslashGTSdAD/\backslashGTSdAU/\backslashGTSmO/\backslashGTSnDAD/\backslashGTSnDAU/]\}}`.

Oriented graph:

$[\overline{m}^{\circ} \vee^{\circ m} \vee^{\circ m}]$  is obtained with

`\mathrel{\{[\backslashGTSdAD/\backslashGTSnDAU/\backslashGTSmO/\backslashGTSnDAD/\backslashGTSdAU/\backslashGTSmO/\backslashGTSnDAD/\backslashGTSnDAU/]\}}`.

Prefix notation:

$\exists u, v, u [\overline{m}^{\circ} \vee^{\circ m} v] \Leftrightarrow \exists u, v, u [\overline{m}^{\circ} \vee^{\circ m} v]$ .

$\exists v, e, v [\overline{m}^{\circ} \vee^{\circ m} e] \Leftrightarrow \exists v, e, v [\overline{m}^{\circ} \vee^{\circ m} e]$ .

$[\overline{m}^{\circ} \vee^{\circ m}]$  is obtained with

`\mathrel{\{[\backslashGTSmO/\backslashGTSxa[mC=red,d]\backslashGTSxa[mC=green,d]\backslashGTSxa[mC=blue,u]\}}`.

$[\overline{m}^{\circ} \vee^{\circ m}]$  is obtained with

`\mathrel{\{[\backslashGTSmO/\backslashGTSxi[mC=red,i]\backslashGTSxi[mC=green,i]\backslashGTSxi[mC=blue,o]\}}`.

Apart from these symbol macros, versatile macros to call them all are available with options:

`\GTSxa[]`, `\GTSxadj[]`, or `\GTSxadjacency[]` for adjacency symbols,

`\GTSxi[]`, `\GTSxinc[]`, or `\GTSxincidence[]` for incidence symbols,

`\GTSxml[]`, or `\GTSxmetallogic[]` for meta-logic symbols,

`\GTSx[]{}`  for all symbols

`(\GTSx[]{} = \GTSxa[], \GTSx[]{}i = \GTSxi[], \GTSx[]{}m = \GTSxml[])`.

See Detailed usage section for full syntax of these macros.